IMPORTANT INSTRUCTIONS :

- **Pattern of the Entrance Examination**:-
  Paper containing 180 objective type questions ,from Biology, Physics and Chemistry
- **Use Blue/Black Ball Point Pen only** to darken the appropriate circle. Answers marked with pencil would not be evaluated.
- Each item carries **4marks**. For each correct response the candidate will get 4 marks. For each incorrect response **1mark** will be deducted from the total score

01. One of the following was the contribution by Ernst Mayr
  (1) Made the origin of species the central question of Biology
  (2) Made the origin of species diversity the central question of evolutionary biology
  (3) Published the book titled Anatomy of Seed Plants.
  (4) Discovered triple helical structure of collagen.

02. Red dinoflagellate among the following is
  (1) *Gonyaulax*
  (2) *Trichodesmium*
  (3) *Polysiphonia*
  (4) *Sequoia*

03. Cell wall is with cellulose and algin in
  (1) Green algae
  (2) Red algae
  (3) Brown algae
  (4) Blue – green algae

04. Statement SI: Asexual spores are endogenously produced in *Rhizopus*
  Statement SII : Sexual spores (Meiospores) are exogenously produced in *Aspergillus*
  (1) Both SI & SII are correct
  (2) Both SI & SII are wrong
  (3) Only SI is correct
  (4) Only SII is correct

05. The location of region of meristematic activity in a typical root is
  (1) Few millimeters above the region of maturation
  (2) Few millimeters below the region of maturation
  (3) Proximal to the region of elongation
  (4) Few millimeters above the thimble – like root cap

06. The peptide hormone which plays a significant role in calcium balance in body
  (1) Aldosterone
  (2) Thyrocalcitonin
  (3) Parathyroid hormone
  (4) Vasopressin

07. Which of the following pairs of animals are similar to each other pertaining to the features stated against them?
  (1) *Macropus* and *Neophron* – Oviparous homeotherms
  (2) *Adamsia* and *Ctenoplana* – Marine, diploblastic, radially symmetrical
  (3) *Ichthyosaurus* and *Bufo* – three chambered heart and scaleless skin
  (4) *Echinus* and *Pinctada* – Spiny body and water vascular system

08. Mid brain is characterised by a pair of optic lobes in
(1) *Rana*
(2) *Delphinus*
(3) *Homo sapiens*
(4) Platypus

09. The enzyme which digests milk sugar is secreted by
(1) Gastric glands
(2) Pancreas
(3) Salivary glands
(4) Intestinal glands

10. Immunodeficiency in HIV infected persons is due to destruction of
(1) B- lymphocytes
(2) Macrophages
(3) T- lymphocytes
(4) NK cells

11. Phloem element absent in most of the monocots is
(1) Phloem parenchyma only
(2) Sieve tube elements only
(3) Companion cells only
(4) Both 2 and 3

12. Identify the gymnosperms that show symbiotic association
(1) *Glomus* and *Nostoc*
(2) *Cycas* and *Pinus*
(3) *Arachis* and *Pisum*
(4) *Casuarina* and *Alnus*

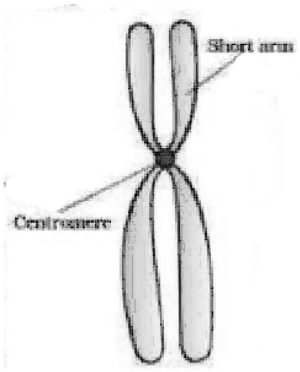
13. The ribosomes of prokaryotes (70S type) is
(1) 30S & 40S
(2) 40S & 60S
(3) 30S & 50S
(4) 40S & 50S

14. Match the following

List – I	List – II
A. Xylem fibres	i) Flax
B. Xylem parenchyma	ii) Obliterated central lumen
C. Phloem parenchyma	iii) Stores food materials, resins, latex
D. Phloem fibres	iv) Stores food materials like starch and other materials like tannins

	A	B	C	D
(1)	ii	iii	i	iv
(2)	i	ii	iii	iv
(3)	ii	iv	iii	i
(4)	iv	iii	ii	i

15. Identify the given below diagram



(1) Sub metacentric
(2) Metacentric
(3) Acrocentric
(4) Telocentric

16. The possible genotypes and phenotypes of blood groups in the progeny of a couple with heterozygous 'A' blood group and homozygous 'B' blood group
(1) 4 phenotypes, 4 genotypes
(2) 1 phenotypes, 2 genotypes
(3) 2 phenotypes, 3 genotypes
(4) 2 phenotypes, 2 genotypes

17. Sexually transmitted infection targeting the liver is
    (1) Genital herpes
    (2) Hepatitis – B
    (3) Trichomoniasis
    (4) Syphilis

18. Choose incorrect pair with reference to joints
    (1) Parietal and frontal bones – Sutures
    (2) Atlas and axis – Condyloid joint
    (3) Humerus and scapula – Ball and socket joint
    (4) Carpal and metacarpal of thumb – Saddle joint

19. Which of the following sets of examples is not convergent evolution ?
    (1) Eye of Octopus and eye of shark
    (2) Flippers of whale and wing of bat
    (3) Sugar glider and flying squirrel
    (4) Sweet potato and potato

20. World summit on sustainable development in 2002 was held in
    (1) South Africa
    (2) Argentina
    (3) Brazil
    (4) Montreal

21. Stomatal apparatus constitutes how many of the following
    i) Guard cells
    ii) Sub stomatal cavity
    iii) Stomatal aperture
    iv) Subsidary cells
    (1) i, ii and iii
    (2) ii, iii and iv
    (3) i, iii and iv
    (4) i, ii and iv

22. In *Amoeba* the contractile vacuole is essential for
    (1) Sexual reproduction
    (2) Storage of food
    (3) Asexual reporudction
    (4) Excretion

23. The correct statement among the following is
    (1) Metaphase of mitosis involves division / split of centromere
    (2) Anaphase of mitosis involves spindle fibres attachment to kinetochore
    (3) Telophase of mitosis involves disappearance of nuclear membrane and endoplasmic reticulum
    (4) Metaphase of mitosis involves attachment of spindle fibres to kinetochores and brining the chromosomes to the equator

24. Most of the organelles in a mitocyte duplicate during
    (1) G_1 - phase
    (2) S - phase
    (3) G_2 - phase
    (4) Prophase

25. Incorrect statement among the following is
    (1) Living state is a non – equilibrium steady – state
    (2) Important form of energy currency in living systems is in ATP
    (3) Living state is a constant effort to reach equilibrium
    (4) Living organisms cannot afford to reach equilibrium

26. Choose correct set from the following.

(a) Globulins in plasma are involved in defence of body

(b) Multipolar neurons are found in cerebral medulla

(c) Neutrophils are most abundant MNP

(d) Smooth muscles are seen in tunica media of veins

(1) a, b

(2) b, c

(3) a, c

(4) a, d

27. Match the items in column – I with those in column – II and select the correct option given below.

Column – I	Column – II
A) Green revolution	I) UV – B radiation
B) Snow blindness	II) Leukemia
C) Nitrogen oxides	III) Ciliotoxins
D) Jhum cultivation	IV) Eutrophication
E) Radioactive wastes	V) Deforestation

(1) A – IV, B – I, C – III, D – V, E – II

(2) A –IV, B – I, C – II, D – V, E – III

(3) A – IV, B – II, C – I, D – V, E – III

(4) A – V, B – I, C – III, D – IV, E – II

28. 'Fimbriae' are associated with this part of reproductive system

(1) ampullla

(2) infundibulum

(3) areola

(4) epididymis

29. Fresh water fish

(1) Pomfret

(2) Sardine

(3) Mackerel

(4) Common carp

30. The extinction of few invertebrate species in the absence of *Pisaster* is due to

(1) Predation

(2) Competition

(3) Parasitism

(4) Amensalism

31. Evolution of the C_4 photosynthetic system is one of the strategies for

(1) Maximising the availability of CO_2 while minimizing water loss

(2) Maximising water loss while minimizing the availability of CO_2

(3) Prevent CO_2 uptake

(4) Prevent O_2 release

32. Toxic secondary metabolite is

(1) Codeine

(2) Ricin

(3) Vinblastin

(4) Anthocyanin

33. During amides formation which part of aminoacid is replaced by another NH_2^- radical.

(1) Hydrogen part

(2) R. group

(3) Hydroxyl part

(4) NH_2^- part

34. Which macronutrient is an immobile element

(1) Calcium

(2) Magnesium

(3) Potassium

(4) Nitrogen

35. The protein part of the conjugated enzyme (holoenzyme) is known as
    (1) Metal ion Cofactor
    (2) Co – enzyme
    (3) Apoenzyme
    (4) Prosthetic group
36. JG cells secrete
    (1) Rennin
    (2) Angiotensinogen
    (3) Atrial natriuretic peptide
    (4) Erythropoietin
37. Radula is found in
    (1) *Aplysia*
    (2) *Ascidia*
    (3) *Antedon*
    (4) *Adamsia*
38. The total amount of air expelled by a person during forceful expiration (after normal inspiration) is approximately
    (1) 3.5 L
    (2) 1.5 L
    (3) 2.3L
    (4) 4.5 L
39. The male accessory gland in *Periplaneta* is
    (1) Prostate gland
    (2) Collaterial gland
    (3) Mushroom gland
    (4) Cowper's gland
40. The protein in the photosensitive pigments of rods is
    (1) Retinal
    (2) Opsin
    (3) Globin
    (4) Troponin
41. Who among the following showed for the first time that green part of the plant only could release oxygen in the presence of light ?
    (1) Joseph Priestly
    (2) T.W. Engelmann
    (3) Julius von Sachs
    (4) Ingenhousz
42. Yeasts poison themselves when the concentration of alcohol reaches
    (1) About 13%
    (2) Above 31%
    (3) About 3%
    (4) Above 30%
43. Plant growth regulators that are antagonistic to each other with regard to seed dormancy
    (1) Auxins - cytokinins
    (2) ABA and GAs
    (3) Ethylene and cytokinins
    (4) All the above
44. Incorrect statement for Asexual reproduction is
    (1) Offspring formed from single parent
    (2) Offspring formed from single parent with (or) without gametes formation
    (3) Offspring produced when two parents participate and involve fusion of gametes
    (4) Offspring produced are identical to one another also to parent
45. Fertilization is external in
    (1) Amphibians
    (2) Majority aquatic algae
    (3) Fishes
    (4) More than one options are correct

46. Choose incorrect pair with reference to excretory organs

(1) *Balaenoptera* – Proboscis gland

(2) *Locusta* – Malpighian tubules

(3) *Fasciola* – Flame cells

(4) *Nereis* – Nephridia

47. Choose incorrect statement from the following.

(1) Cerebral aqueduct passes through cerebrum.

(2) Human kidneys excrete 25 – 30 gms of urea per day

(3) Calcium ions are essential for blood clotting and muscle contraction

(4) Deficiency of iodine in diet can cause goitre

48. Sometimes doctors inject a hormone to induce delivery after full term pregnancy. Which hormone is injected? How does it induce delivery?

(1) Mifepristone to induce abortion by stimulating uterine contractions

(2) Relaxin to induce labour pains

(3) Cortisols to suppress foetal rejection

(4) Oxytocin to stimulate uterine contractions.

49. The first meiotic division during gametogenesis of human produces

(1) Spermatid

(2) Second polar body

(3) Secondary spermatocyte

(4) Primary oocyte

50. The ex situ conservation methods of biodiversity for animals

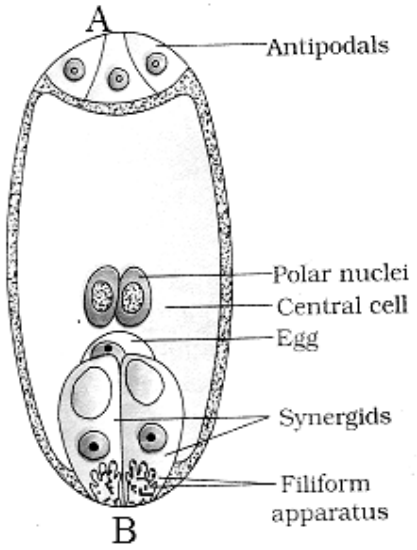
(1) Safari park and sperm banks

(2) Sacred grove and zoological park

(3) Botanical garden and seed banks

(4) Tissue culture method and pollen banks

51. Identify A and B from the diagram below



(1) A – micropylar end and B – chalazal end

(2) A – chalazal end and B – micropylar end

(3) A and B are posterior ends

(4) A and B are anterior ends

52. Hydrophily is seen in

(1) 13 species of hydrophytes only

(2) 30 genera of hydrophytes only

(3) Majority of hydrophytes

(4) Majority of xerophytes

53. Apomictic seeds are common in

(1) Asteraceae

(2) Grasses

(3) *Cycas*

(4) Both (1) and (2)

54. How many of the following represent 1 : 2 : 1 ratios

a. Genotypes of F_2 progeny of monohybrid cross

b. Genotypes of F_2 progeny of dihybrid cross

c. Genotypes of progeny of monohybrid test cross

d. Phenotypes of F_2 progeny of monohybrid cross

e. Phenotypes and genotypes in F_2 progeny involving incomplete dominance

f. Phenotypes and genotypes in F_2 progeny involving co - dominance

(1) 3

(2) 4

(3) 5

(4) 6

55. The term linkage was coined by

(1) Mendel

(2) Sutton and Boveri

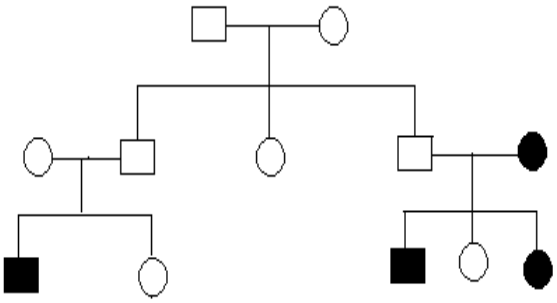
(3) Punnet

(4) Morgan

56. Choose the correct matching of the given items with their corresponding group.

	Items	Group
(1)	Malleus, incus, scapula	Ear ossicles
(2)	Medulla, pons, cerebellum	Brain stem
(3)	Frontal, ethmoid, sphenoid	Cranial bones
(4)	Fovea, macula, optic disc	Retina of eye

57. Identify the genetic trait from the given pedigree analysis by observing the inheritance pattern.



(1) Haemophilia

(2) Sickle – cell anaemia

(3) Myotonic dystrophy

(4) Incontinentia pigmenti

58. Choose correct combination with reference to human evolution

(1) *Australopithecines* – hunted with stone weapons about 1.5 mya

(2) *Homo erectus* – first hominid with brain capacity of 650 cc and ate meat

(3) Neanderthal man – lived in central Asia and buried their dead

(4) *Homo sapiens* – developed prehistoric cave art about 1,00,000 years ago

59. Which of the following is not a vector borne disease?

(1) Dengue

(2) Filariasis

(3) Delhi boils

(4) Diphtheria

60. Match the contents in column I with column II and choose the correct option.

Column I	Column II
A) Parkinson's disease	I) DDT pollution
B) Biomagnification	II) Degeneration of thymus
C) Immunodeficiency	III) Low level of estrogen
D) Diabetes mellitus	IV) Hyper glycemia
E) Osteoporosis	V) Loss of dopamine

(1) A – V, B – I, C – IV, D – II, E – III

(2) A – V, B – I, C – II, D – IV, E – III

(3) A – V, B – II, C – I, D – IV, E – III

(4) A – V, B – I, C – III, D – IV, E – II

61. Alfred Hershey and Martha Chase provided unequivocal evidence to prove genetic material as DNA by using radio isotopes, namely

(1) S^{32}, P^{35}

(2) S^{35}, P^{32}

(3) C^{14}, N^{15}

(4) C^{15}, N^{14}

62. Match the following regarding nucleotides in DNA

List – I	List – II
A. Diploid human cell DNA	i) 5386 nucleotides
B. *E.Coli* DNA	ii) 48502 base pairs (bp)
C. Bacteriophage lambda DNA	iii) 4.6×10^{6} base pairs
D. Bacteriophage ϕx174 phage DNA	iv) 6.6×10^{9} base pairs

A B C D

(1) iv iii ii i

(2) iv iii i ii

(3) i ii iii iv

(4) iii iv ii i

63. The substrates for replication of DNA with dual role

(1) Ribonucleoside triphosphates

(2) Any nucleoside triphosphates

(3) Deoxyribonucleoside triphoshates

(4) DNA polymerases

64. UTRs in a translation unit are locted at

(1) Before stop codon and after start codon

(2) Before and after stop codon as well as start codon

(3) Before start codon and after stop codon

(4) Between start codon and stop codon

65. Inducer of lac operon is

(1) Lactose or allolactose

(2) β – galactosidase or permease

(3) Transacetylase or repressor protein

(4) All the above

66. Which of the following factors favour dissociation of oxygen from oxyhaemoglobin?

(a) High pH

(b) High pCO_2

(c) High temperature

(d) High pO_2

(1) a, b, c

(2) b, c, d

(3) a, b only

(4) b, c only

67. Autoimmune disorder causing paralysis of muscle

(1) Tetany

(2) Muscular dystrophy

(3) Myasthenia gravis

(4) Sarcoma

68. Increase in ventricular pressure of heart causes

(1) Closure of A-V valves

(2) Closure of semilunar valves

(3) Opening of A–V valves

(4) Ventricular filling

69. Hag fish, dog fish and devil fish are similar in possessing the following features

(1) Marine animals with pharyngeal gills for respiration

(2) Bilateral symmetry and true coelom

(3) Internal fertilisation and indirect development

(4) Metamerism and closed circulation

70. Catalytic converters are used in automobiles to reduce the pollution of

(1) Sulphur dioxide

(2) Carbon monoxide

(3) Particulate matter

(4) CFC

71. The palindromic sequence for *Hind* – II enzyme is

(1) $\frac{^{5^1}GTP_yP_uAC^{3^1}}{^{3^1}CAP_uP_yTG^{5^1}}$

(2) $\frac{^{5^1}GAATTC^{3^1}}{^{3^1}CTTAAG^{5^1}}$

(3) $\frac{^{5^1}AAGCTT^{3^1}}{^{3^1}TTCGAA^{5^1}}$

(4) $\frac{^{5^1}AGCT^{3^1}}{^{3^1}TCGA^{5^1}}$

72. In genetic engineering T_{aq} polymerase is used for

(1) Gel electrophoresis for isolating DNA

(2) PCR method for extension of primers

(3) PCR method for annealing with primers

(4) Gel electrophoresis for visualization

73. Toxin proteins of *Bacillus thuringiensis* have killing effect on dipterans like

(1) Mosquitoes and beetles

(2) Mosquitoes and army worm

(3) Budworm and beetles

(4) Flies and mosquitoes

74. Proteins encoded by which genes are resistant to corn borer

(1) CryIAc

(2) CryIIAb

(3) CryIAb

(4) 1 and 2

75. Amount of carbon fixed annually, round the globe is

(1) $49\times10^{13}\,kg$

(2) $71\times10^{13}\,kg$

(3) $4\times10^{13}\,kg$

(4) $3\times10^{14}\,kg$

76. Statement A: Spleen is grave yard of RBC.

Statement B: Lymph borne antigens are filtered by spleen to activate immunocompetent lymphocytes.

(1) Both statement A and statement B are correct

(2) Both statement A and statement B are incorrect

(3) Statement A is correct but statement B is incorrect

(4) Statement A is incorrect but statement B is correct

77. Choose incorrect statement about glucocorticoids

(1) Produce inflammatory response

(2) Stimulate erythropoiesis

(3) Inhibit protein synthesis

(4) Elevate blood glucose level

78. The various trophic levels of a food chain are given below.

Big tree $\rightarrow$ Insects $\rightarrow$ Insectivorous birds $\rightarrow$ Larger birds.

Which of the following ecological pyramid explains the relationship between all trophic levels?

(1) Inverted pyramid of numbers

(2) Inverted pyramid of biomass

(3) Spindle shaped pyramid of biomass

(4) Upright pyramid of biomass

79. The epithelium lining alveoli of lungs is similar to

(1) endometrium of uterus

(2) endosteum of bone

(3) endothelium of blood vessels

(4) endomysium of muscle fibre

80. GIFT is suggested to an infertile woman who can

(1) produce ova, but cannot conceive

(2) not produce ova and cannot conceive

(3) not produce ova, but can conceive

(4) produce ova and can conceive

81. Statins obtained from yeast *Monascus* purpureus lowers blood cholesterol by acting as

(1) Non – competitive inhibitor

(2) Co - factor

(3) Prosthetic group

(4) Competitive inhibitor

82. Swiss cheese with large holes is obtained by employing

(1) *Saccharomyces*

(2) *Trichoderma*

(3) *Propionibacterium*

(4) *Monascus*

83. In sewage treatment biogas is obtained from

(1) Primary sludge in secondary treatment

(2) Activated sludge from settling tank

(3) Anaerobic sludge digester during secondary treatment

(4) Aerating tank in primary treatment

84. Which of the following is a bacterial disease

(1) Late blight of potato

(2) Black rot of crucifers

(3) Turnip mosaic

(4) Red rot of sugar cane

85. Cybrids (cytoplasmic hybrids) are obtained by fusion of
(1) Nuclei of two cells and cytoplasm from one cell
(2) Nuclei and cytoplasm of two cells
(3) Nucleus of one and cytoplasm of two cells
(4) Nuclei and cytoplasm of three cells

86. Statement A: Small animals like shrews and humming birds are not seen in polar regions.
Statement B: They are thermo conformers.
(1) Both statement A and statement B are correct
(2) Both statement A and statement B are incorrect
(3) Statement A is correct but statement B is incorrect
(4) Statement A is incorrect but statement B is correct

87. Choose incorrect combination.

	Chemical substance	Secreted by	Function
(1)	Nucleosidase	Crypts of Lieberkhun	Produces sugar and nitrogen bases by digestion
(2)	Melatonin	Pituitary	Regulates circadian rhythms
(3)	ADH	Hypo thalamus	Water reabsorption in DCT
(4)	Lysozyme	Paneth cells	Kills bacteria

88. The method of sex determination in grass hopper
(1) X X – XY
(2) X X – XY
(3) Z W – Z Z
(4) haplodiploidy

89. If the recessive genotype frequency is 0.25 in a Hardy – Weinberg population, the genotype frequency of heterozygous individuals will be
(1) 0.5
(2) 0.75
(3) 0.37
(4) 0.25

90. Human male with trisomy of allosomes is related to
(1) Down syndrome
(2) Turner syndrome
(3) Edward's syndrome
(4) Klinefelter syndrome

91. The potential energy of a particle varies with distance x from a fixed orgin as $V = \frac{A\sqrt{x}}{x+B}$ where A and B are constants. The dimensions of AB are
(1) $M^1L^{7/2}T^{-2}$
(2) $M^1L^2T^{-2}$
(3) $M^{3/2}L^{5/2}T^{-2}$
(4) $M^1L^{5/2}T^{-2}$

92. Person A moving with a constant velocity of 20m/s wishes to catch person B who starts from rest and moving with constant acceleration of $2ms^{-2}$ in same direction 100m ahead of him. Determine time taken by person A to catch B
(1) 10 s
(2) 20 s
(3) 30 s
(4) 40 s

93. A person walking at 4m/s finds rain drops falling slant wise in to his face with a speed of 4m/s at an angle of 30° with the vertical. The actual speed of the rain drops is

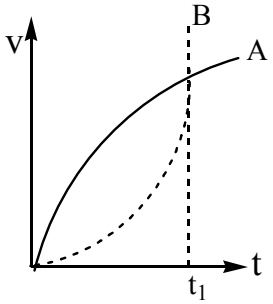
(1) 4m/s

(2) 8m/s

(3) 6m/s

(4) 5m/s

94. At t = 0, two trucks A and B were at same point on the road. They are moving along parallel lines. Motion of trucks A and B are represented by bold and dotted lines respectively. At time $t = t_1$



(1) Truck B is ahead of truck A

(2) Acceleration of trucks A and B are equal

(3) Speed of truck A decreases and speed of truck B increases

(4) Truck A is ahead of truck B

95. A projectile is thrown, from the origin along X-Y plane to have maximum possible horizontal range of 200 m. The position vector of the projectile when its speed is minimum is (take Y on the vertical axis)

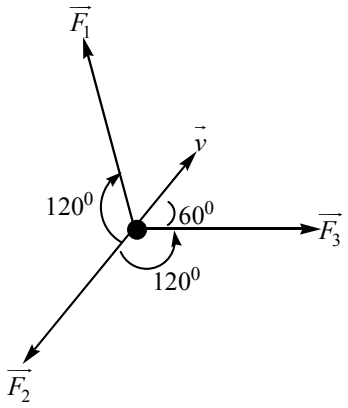
(1) $\left(100\hat{i}+100\hat{j}\right)m$

(2) $\left(100\hat{i}+200\hat{j}\right)m$

(3) $\left(100\hat{i}+50\hat{j}\right)m$

(4) $\left(200\hat{i}+100\hat{j}\right)m$

96. A particle is moving in a plane with velocity 'v' as shown. If it is now acted upon by forces $\vec{F_1}, \vec{F_2}$ and $\vec{F_3}$ of magnitude 10N, 10N and 15N respectively in the same plane as shown then



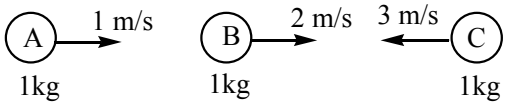
(1) its velocity changes only in magnitude

(2) its velocity changes only in direction

(3) its velocity changes in both magnitude and direction

(4) its velocity remains constant

97. Three bodies are moving as shown below. The total kinetic energy of the system of three masses in the frame of reference of 'B' is



1) 13 J

(2) 25 J

(3) 20.5 J

(4) 8.5 J

98. An engine is of 10kW power. How much time will it take to lift a mass of 200kg to a height of 40m? ($g=10ms^{-2}$)

(1) 4s

(2) 8s

(3) 5s

(4) 10s

99. In a circular motion of a particle, the tangential acceleration of the particle is given by $a_t = 9m/s^2$. The radius of the circle is 4m. The particle was initially at rest. Time after which acceleration of the particle makes an angle of 45^0 with the radial acceleration is

(1) $\frac{1}{3}s$

(2) $\frac{2}{3}s$

(3) $1s$

(4) $\frac{4}{3}s$

100. A ball of mass 'm' moving at speed 'u' makes a head on collision with an identical ball at rest. The kinetic energy of the balls after the collision is three-fourth of the original. The coefficient of restitution (e) is

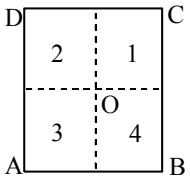
(1) 1

(2) 0

(3) $\frac{1}{2}$

(4) $\frac{1}{\sqrt{2}}$

101. In the given figure four rods AB, BC, CD and DA of same length and of masses 'm', 2m, 3m and 4m respectively arranged in the form of a square. The region in which (numbered 1,2,3,4) the centre of mass of system lies in



(1) region 1

(2) region 2

(3) region 3

(4) region 4

102. A disc at rest at the top of the inclined plane of height 'h' rolls down without slipping and acquires a velocity 'v' on reaching the bottom. If the same disc slides down a smooth incline and acquires the same velocity on reaching the bottom then the height of smooth incline is

(1) h/3

(2) h/2

(3) 2h/3

(4) h

103. A satellite moves around the earth in a circular orbit with speed 'v'. If 'm' is the mass of the satellite then its potential energy is

(1) $-\frac{1}{2}mv^2$

(2) $\frac{1}{2}mv^2$

(3) $-mv^2$

(4) $\frac{3}{2}mv^2$

104. Three moles of an ideal gas 'A' with $\frac{C_P}{C_V} = \frac{4}{3}$ is mixed with two moles of another ideal gas 'B' with $\frac{C_P}{C_V} = \frac{5}{3}$ at constant temperature. The ratio $\frac{C_P}{C_V}$ of the mixture is

(1) 1.5

(2) 1.42

(3) 1.7

(4) 1.3

105. An ideal gas occupies a volume of $2m^3$ at a pressure of 3×10^6 Pa. The energy of the gas is

(1) 3×10^2

(2) 10^8 J

(3) 6×10^4 J

(4) 9×10^6 J

106. A block of mass 'm' and surface area A just begins to slide down an incline when the angle of inclination is 30^0. Keeping the mass of the block same, if the surface area is doubled, without any change in surface finish then the inclination of the plane at which the block starts sliding will be

(1) 60^0

(2) 30^0

(3) 15^0

(4) 45^0

107. If a body is performing SHM, then its

A) average kinetic energy per cycle is equal to half of its maximum kinetic energy

B) mean speed over a complete cycle is equal to $\frac{2}{\pi}$ times of its maximum speed

(1) Only A is correct

(2) Only B is correct

(3) Both A and B are correct

(4) Both A and B are wrong

108. A simple pendulum with a metal bob has a time period 'T'. Now the bob is immersed in a liquid which is non viscous. Thus the time period is '3T'. Then the ratio of densities of metal bob and that of liquid is

(1) 8 : 9

(2) 9 : 8

(3) 1 : 9

(4) 9 : 1

109. A tank containing water has an orifice of in one vertical side. It is 10m below the surface level of water in tank. Then velocity of efflux from that small orifice is

(1) $7ms^{-1}$

(2) $14ms^{-1}$

(3) $9.8ms^{-1}$

(4) $3.5ms^{-1}$

110. Bulk modulus of a material, whose Poison's ratio equal to 0.5 is

(1) Zero

(2) infinity

(3) of the order of 10^{11} Pa

(4) of the order of 10^4 Pa

111. Oil rises up in the wick of diya (lamp) because

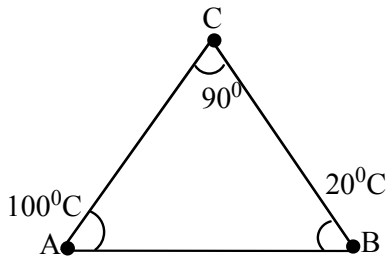
(1) of high surface tension of oil

(2) of capillaries formed in the wick

(3) angle of contact between oil and wick is obtuse

(4) cohesive forces between oil molecules predominates adhesive forces

112. Three rods of identical cross-section and of same material form the sides of right angled isosceles triangle as shown. If the points A and B are maintained at 100^0C and 20^0C, then the steady state temperature of point C, will be



(1) 60^0C

(2) 90^0C

(3) 30^0C

(4) 40^0C

113. A given ray of light suffers minimum deviation in an equilateral prism P. If refractive index increases slightly, then the ray will now suffer

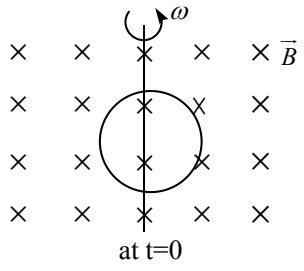
(1) greater deviation

(2) no deviation

(3) same deviation as before

(4) less deviation

114. A ring is rotated about a diametric axis in a uniform magnetic field perpendicular to the plane of the ring with a time period 10 s. If initially the plane of the ring is perpendicular to the magnetic field then the instant of time at which EMF will be maximum & minimum respectively for the first time is



(1) 10 s, 5 s

(2) 5 s, 7.5 s

(3) 2.5 s, 7.5 s

(4) 2.5 s, 5 s

115. The focal length of convex lens is 10 cm. Find its magnifying power when it is used as a magnifying glass to form the image at near point

(1) 1.5

(2) 2.5

(3) 2.5

(4) 3.5

116. Which of the following statements is wrong?

(a) Yellow light is used in fog lamps for vehicles as it is scattered most by fog particles

(b) Red light is used as danger signal as it is scattered least by atmospheric particles

(c) In primary rainbow violet appears on the top and red at the bottom

(1) only a

(2) only b

(3) both a and c

(4) both b and c

117. An unpolarised light is incident on a plate of refractive index $\sqrt{3}$ and the reflected light is found to be completely plane polarised . The angles of incidence and refraction are respectively.

(1) $\sin^{-1}\left(\frac{1}{\sqrt{3}}\right), 45^0$

(2) $30^0, 60^0$

(3) $60^0, 30^0$

(4) $\tan^{-1}\left(\frac{\sqrt{3}}{2}\right), 30^0$

118. A young's double slit experiment uses two monochromatic point sources. The shape of interference fringes formed on the screen parallel to plane of the slits is

(1) Hyperbola

(2) Parabola

(3) Straight line

(4) Circle

119. At an instant, two electrons enter normally with same speed v, one into region of uniform electric field and other into uniform magnetic field. After some time, if de Broglie wavelength of two electrons respectively are λ_1 and λ_2 then

(1) $\lambda_1 = \lambda_2$

(2) $\lambda_1 > \lambda_2$

(3) $\lambda_1 < \lambda_2$

(4) $\lambda_1 = 2\lambda_2$

120. The force (F) acting on the electron in hydrogen atom depends on the principle quantum number 'n' as

(1) $F\alpha\frac{1}{n^4}$

(2) $F\alpha n^2$

(3) $F\alpha n^4$

(4) $F\alpha\frac{1}{n^2}$

121. The half-life of a radioactive nucleus is 50 days. The time interval (t_2-t_1) between the time t_2 when 2/3 of it has decayed and the time t_1 when 1/3 of it had decayed is

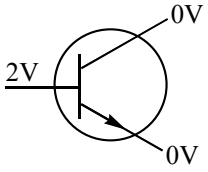
(1) 60 days

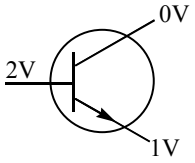
(2) 15 days

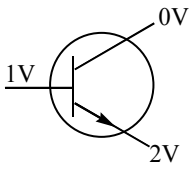
(3) 30 days

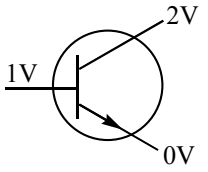
(4) 50 days

122. In which of the following cases, the transistor is operating in the active region?

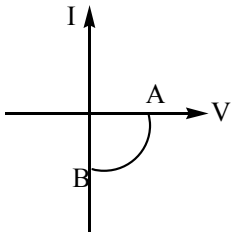
(1) 

(2) 

(3) 

(4) 

123. The given graph represents V-I characteristic for a semiconductor device. Which of the following statements is correct?



(1) It is V-I characteristic for solar cell where, point A represents open circuit voltage and point B short circuit current

(2) It is for a solar cell and point A and B represent open circuit voltage and current, respectively

(3) It is for photodiode and point A and B represent open circuit voltage and current, respectively

(4) It is for a LED and points A and B represent open circuit voltage and short circuit current, respectively

124. An LCR series circuit is compared with a damped oscillator of mass 'm' with force constant K and damping coefficient 'b'. The correct matching of quantities in damped oscillator analogous with those in LCR circuit is

(1) $L \to m, C \to \frac{1}{k}, R \to b$

(2) $L \to m, C \to k, R \to b$

(3) $L \to k, C \to b, R \to m$

(4) $L \to \frac{1}{m}, C \to \frac{1}{k}, R \to \frac{1}{b}$

125. Two infinite plane sheets have uniform surface charge density $+\sigma$ Cm^{-2}. If they are parallel to each other then the magnitude of electric field strength at a point in between the plane sheets of charge is

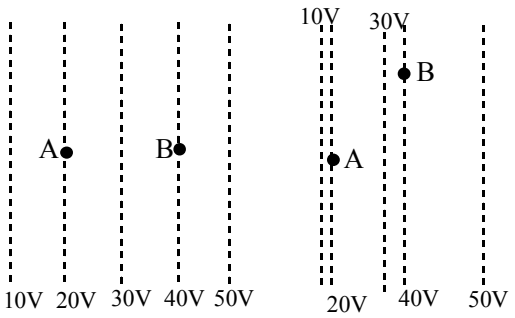
(1) zero

(2) $\frac{\sigma}{\varepsilon_0}$

(3) $\frac{2\sigma}{\varepsilon_0}$

(4) $\frac{3\sigma}{4\varepsilon_0}$

126. Figure shows some equipotential lines distributed in space. A charged particle is moved from point A to point B



(1) The work done in field I is greater

(2) The work done in field I is smaller

(3) The work done is the same in field I and field II

4) The work done is zero in field I and is positive in field II

127. In a potentiometer experiment, the galvanometer shows no deflection when a cell is connected across 60cm of the potentiometer wire. If the cell is shunted by a resistance of 6Ω, the balance is obtained across 50cm of the wire. The internal resistance of the cell is

(1) 0.5Ω

(2) 0.6Ω

(3) 1.2Ω

(4) 1.5Ω

128. A dip circle is placed in geographic meridian at a place where dip and declination are δ and D respectively. The dip measured by dip circle is

(1) δ

(2) $\tan^{-1}\left(\frac{\tan\delta}{\sin D}\right)$

(3) $\tan^{-1}\left(\frac{\cos D}{\tan\delta}\right)$

(4) $\tan^{-1}\left(\frac{\tan\delta}{\cos D}\right)$

129. If a long hollow copper pipe carries a direct current, the magnetic field associated with the current will be

(1) Only inside the pipe

(2) Only outside the pipe

(3) Both inside and outside the pipe

(4) Neither inside nor outside the pipe

130. An electron orbiting around a nucleus has angular momentum L. The magnetic field produced by the electron at the centre of the orbit can be expressed as (Terms have their usual meaning)

(1) $\left(\frac{\mu_0 e}{8\pi mr^3}\right)L$

(2) $\left(\frac{\mu_0 e}{4\pi mr^3}\right)L$

(3) $\left(\frac{\mu_0 e}{\pi mr^3}\right)L$

(4) $\left(\frac{e}{4\pi \in_0 mr^3}\right)L$

131. Ozone layer blocks

(1) Infrared radiation

(2) Microwaves

(3) Ultraviolet radiation

(4) Visible radiation

132. Two circular coils 1 and 2 having same number of turns are made from the same wire but the radius of the 1st coil is twice that of the 2nd coil. The ratio of potential difference applied across them so that the magnetic field at their centres is the same is

(1) 3

(2) 2

(3) 6

(4) 4

133. On the perpendicular bisector of an electric dipole the electric field intensity 'E' and potential 'V' are

(1) E = O; V = O

(2) $E \neq O$; $V = O$

(3) $E \neq O$; $V \neq O$

(4) $E = O$; $V \neq O$

134. Two wires made of same material having same length and of radii 'r' and 2r respectively are welded together end to end. The combination is used as a sonometer wire and is kept under a tension T. When stationary waves are set up in the wires then joint is a node. The ratio of number of loops formed in the wires, is

(1) 2 : 3

(2) 1 : 1

(3) 1 : 2

(4) 1 : 4

135. The wavelength of the spectral line coming from a star is changed from $6000A^0$ to $6001A^0$. The velocity of the star with respect to the earth is

(1) $2.5\times10^4 ms^{-1}$ approach

(2) $2.5\times10^4 ms^{-1}$ recession

(3) $5\times10^4 ms^{-1}$ approach

(4) $5\times10^4 ms^{-1}$ recession

136. The equilibrium pressure of $NH_4CN(s) \leftrightarrow NH_3(g) + HCN(g)$ is 20 atm. Then K_p is

(1) 100 atm^2

(2) 40 atm^2

(3) 85 atm^3

(4) 75 atm

137. Which among the following is incorrect expression of Heisenberg's uncertainty principle ?

(1) $\Delta x \times \Delta P \geq \frac{h}{4\pi}$

(2) $\Delta t \times \Delta E \geq \frac{h}{4\pi}$

(3) $\Delta x \times \Delta V \geq \frac{hm}{4\pi}$

(4) $\Delta x \times \Delta \lambda \geq \frac{\lambda^2}{4\pi}$

138. Which of the following is used to prevent heart attacks ?

(1) Codeine

(2) Amoxycillin

(3) Penicillin

(4) Aspirin

139. Which of the following atoms has the highest second ionization energy

(1) Mg

(2) Si

(3) Na

(4) A*l*

140. Zieglar – Natta catalyst is an organometallic compound of which metal

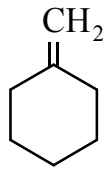
(1) Iron

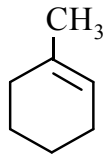
(2) Zirconium

(3) Rhodium

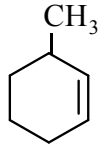
(4) Titanium

141. In the reaction with HC*l*, an alkene reacts in accordance with the Markovnikov's rule, to give a product 1-chloro-1-methyl cyclo hexane. The possible alkene is

(1) 

(2) 

(3) 1 and 2

(4) 

142. Which oxide of nitrogen is not a common pollutant introduced into the atmosphere both due to natural and human activity ?

(1) N_2O

(2) NO

(3) NO_2

(4) N_2O_5

143. Which among the following set is incorrect ?

(1) Increasing order of bond order:

$O_2^{-2} < O_2 < O_2^{+1}$

(2) Increasing order of bond angle :

$SnCl_4 < BCl_3 < BeCl_2$

(3) Increasing order of dipole moment :

$NH_3 < NF_3$

(4) Increasing order of C-C bond length :

Ethyne < Ethene < Benzene < Ethane.

144. $[NiCl_4]^{2-}$ and $[Ni(CN)_4]^{2-}$ show similarity in

(1) Geometry

(2) Magnetic nature

(3) Hybridisation of state of Ni

(4) Primary valency of Ni

145. $H_2O(l) \rightarrow H_2O(g)$, at 387K correct signs for $\Delta H,\ \Delta S,\ \Delta G$ respectively are

(1) $+, +, +$

(2) $-, -, -$

(3) $+, +, -$

(4) $-, +, -$

146. Interhalogen compounds are more reactive than the individual halogens except fluorine because

(1) they are prepared by direct combination of halogens

(2) $X-X'$ bond is weaker than $X-X$ or $X'-X'$ bonds

(3) they are thermally more stable than halogen

(4) there is a large difference in their electronegativity

147. In electrophoresis the colloidal particles of As_2S_3 sol

(1) moves towards anode

(2) moves towards cathode

(3) neither towards cathode nor towards anode

(4) both towards cathode and anode

148. The only actinoid element not exhibiting +3 oxidation state in its compounds:

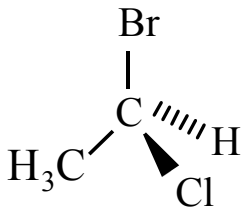
(1) Ac

(2) Th

(3) Cf

(4) Pu

149. The chirality of the compound



(1) R

(2) S

(3) Z

(4) E

150. Solute 'A' is a ternary electrolyte (AB_2) and solute 'B' is a non electrolyte. If 0.1M solution of solute 'B' produces an osmotic pressure of '2p', then 0.05M solution of 'A' at the same temperature will produce an osmotic pressure equal to:

(1) 1.5p

(2) 2p

(3) 3p

(4) P

151. Chlorobenzene on treatment with sodium in dryether gives diphenyl. The name of the reaction is

(1) Fittig reaction

(2) Wurtz Fittig reaction

(3) Gattermann reaction

(4) Wurtz reaction

152. Which among the following undergoes anionic hydrolysis only ?

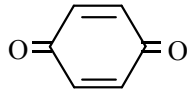
(1) NaCl

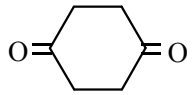
(2) CH_3COONa

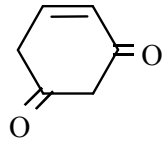
(3) NH_4Cl

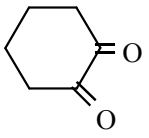
(4) KNO_3

153. Which of the following does not exhibit tautomerism ?

(1) 

(2) 

(3) 

(4) 

154. 80g of oxygen contains as many atoms as in

(1) 10g of hydrogen

(2) 5g of hydrogen

(3) 80g hydrogen

(4) 1g of hydrogen

155. Which among the following pair of species is isostructural ?

(1) PCl_5 & IF_5

(2) H_3O^+ & NCl_3

(3) SO_2 & XeF_2

(4) SO_3 & XeO_3

156. In which pair most efficient packing is present ?

(1) hcp and bcc

(2) hcp and ccp

(3) bcc and ccp

(4) bcc and simple cubic cell

157. The IUPAC name of the Wilkinsons catalyst $[RhCl(P\ Ph_3)_3]$ is

(1) Chlorotris (triphenylphosphine) rhodium (I)

(2) Chlorotris (triphenylphosphine) rhodium (IV)

(3) Chlorotris (triphenylphosphine) rhodium (0)

(4) Chlorotris(triphenylphosphine) rhodium (VI)

158. Pick out the correct statement with respect to $\left[Co\left(C_2O_4\right)_3\right]^{3-}$

(1) It is outer orbital complex, sp^3d^2 hybridized and diamagnetic in nature

(2) It is inner orbital complex, d^2sp^3 hybridized and paramagnetic in nature

(3) It is inner orbital complex, d^2sp^3 hybridized and diamagnetic in nature

(4) It is outer orbital complex, sp^3d^2 hybridized and paramagnetic in nature

159. Wrong match is

(1) Glycyl alanine …..Dipeptide

(2) Alanine ….Neutral amino acid

(3) Insulin …Globular protein

(4) Glycine …. Optically active amino acid

160. The boiling points of water and ethyl alcohol are 373K and 351.3K respectively. The mixture of water and ethyl alcohol at certain composition forms azeotropic mixture and it shows positive deviation from Raoults law. At what minimum temperature the mixture may boil ?

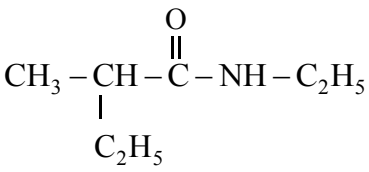
(1) 390K

(2) 351.3K

(3) 373K

(4) 350.15K

161. The IUPAC name of the compound is



(1) N - ethyl - 2 - ethyl propanamide

(2) N - ethyl - 2 - methyl butanamide

(3) N-ethyl-2- methyl propane carboxamide

(4) N - ethyl - 2 - ethyl ethane carboxamide

162. Ion - dipole forces are present between

(1) NaCl & H_2O

(2) SO_2 & H_2O

(3) SF_6 & CH_4

(4) PCl_5 & SO_3

163. Which among CH_4, SiH_4, GeH_4 and SnH_4 is most volatile ?

(1) SnH_4

(2) SiH_4

(3) GeH_4

(4) CH_4

164. A synthetic polyamide prepared by prolonged heating of caprolactum is

(1) Dacron

(2) Nylon-6

(3) Acrilan

(4) Teflon

165. The conjugate acid of HPO_4^{-2} is

(1) HPO_3^{-2}

(2) $H_2PO_4^{-1}$

(3) PO_4^{-3}

(4) H_3PO_4

166. Which of the following slag is produced during extraction of iron ?

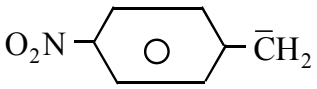
(1) $CaSiO_3$

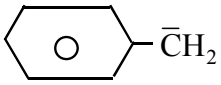
(2) $FeSiO_3$

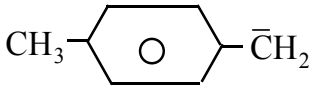
(3) $MgSiO_3$

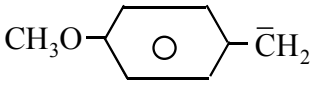
(4) $ZnSiO_3$

167. Which of the following is the most stable carbanion ?

(1) 

(2) 

(3) 

(4) 

168. $X=\left[Li\left(H_2O\right)_n\right]^+$, $Y=\left[K\left(H_2O\right)_n\right]^+$, $Z=\left[Cs\left(H_2O\right)_n\right]^+$, the correct order of size of these hydrated alkali ions

(1) X > Y > Z

(2) Z > Y > X

(3) X = Y = Z

(4) Z > X > Y

169. Incorrectly matched

A: Clemmensen reduction - Na-Hg/HC*l*

B: Mesityl oxide – formed in aldol condensation of acetone

C: Hemiacetal – alkoxy alkane

D: Gattermann – Koch reaction – CO + HC*l* / anhyd A*l*Cl_3

(1) B

(2) C

(3) A

(4) D

170. Which of the following contain neither S – S bond nor S = S bond ?

(1) $S_2O_4^{2-}$

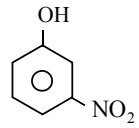
(2) $S_2O_5^{2-}$

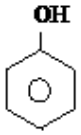
(3) $S_2O_3^{2-}$

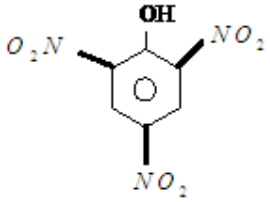
(4) $S_2O_7^{2-}$

171. Most acidic among the following is

(1) CH_3COOH

(2) 

(3) 

(4) 

172. The number of P-O-P and P-O-H bonds present respectively in pyrophosporic acid molecule

(1) 1, 2

(2) 2, 2

(3) 1, 4

(4) 1, 8

173. The following data is observed for the reaction $A + B \rightarrow AB$

[A]	[B]	Rate in M/sec
0.1M	0.1M	2.5×10^{-4}
0.2M	0.1M	5.0×10^{-4}
0.2M	0.2M	10×10^{-4}

The order of reaction is:

(1) 0
(2) 1
(3) 2
(4) 1.5

174. In Kjeldahl's method of estimation of nitrogen, $CuSO_4$ acts as

(1) Oxidising agent
(2) Reducing agent
(3) Catalytic agent
(4) Hydrolysis agent

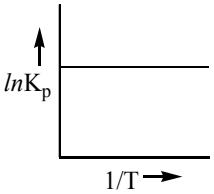
175. According to Fajan's rules, in which among the following polarisability is maximum ?

(1) $CaCl_2$
(2) $CaBr_2$
(3) CaF_2
(4) CaI_2

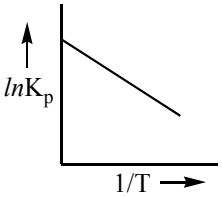
176. The reduction potential of Cu when Copper rod is dipped in 0.1M $CuSO_4$ solution is: (SRP of Cu is +0.34V)

(1) + 0.37V
(2) + 0.28V
(3) - 0.40 V
(4) + 0.31V

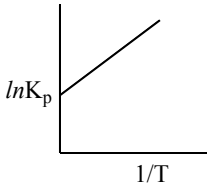
177. For an Exothermic reaction at equilibrium, the valid graphical data is



(1)



(2)

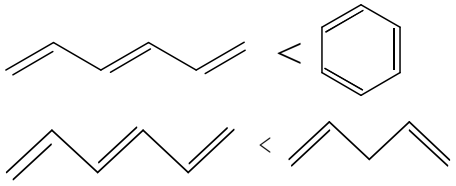


(3)

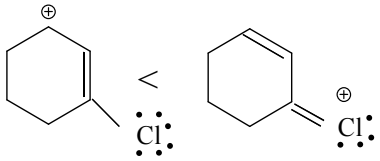
(4) All the above

178. According to the stability which of the following order is correct

a)



b)



c)

(1) a & b only
(2) a & c only
(3) b & c only
(4) a, b, c

179. $\text{Propene} \xrightarrow{HBr} A \xrightarrow{Zn/H^+}$

$B \xrightarrow{\Delta} C + methane$

Incorrect statement is

(1) Intermediate formed in the formation of "A" is $\left(CH_3\right)_2 \overset{+}{C}H$

(2) B is propane

(3) C is ethane

(4) $A \xrightarrow{Na/ether}$ 2, 2, 3, 3-tetramethyl Butane

180. A vessel contains helium and CH_4 in 2 :1 molar ratio at 15 bar pressure. Due to leakage, the mixture of gases starts effusing out. The ratio of initial rates of effusion of He & CH_4 is

(1) 1 : 2
(2) 8 : 1
(3) 1 : 4
(4) 4 : 1

SRI CHAITANYA EDUCATIONAL INSTITUTIONS,INDIA

A.P,TELANGANA,KARNATAKA,TAMILNADU,MAHARASHTRA,DELHI,RANCHI,CHANDIGARH

SR AIIMS S60 **NEET GRAND TEST - 5 KEY** **Date : 25-01-2020**

BOT : 1 – 5, 11 – 15, 21 – 25, 31 – 35, 41 – 45, 51 – 55, 61 – 65, 71 – 75, 81 – 85

ZOO : 6 -10, 16 – 20, 26 – 30, 36 – 40, 46 – 50, 56 – 60, 66 – 70, 76 – 80, 86 – 90

BIOLOGY

1) **2**	2) **1**	3) **3**	4) **3**	5) **4**	6) **3**	7) **2**	8) **1**	9) **4**	10) **3**
11) **1**	12) **2**	13) **3**	14) **3**	15) **1**	16) **4**	17) **2**	18) **2**	19) **2**	20) **1**
21) **3**	22) **4**	23) **4**	24) **1**	25) **3**	26) **4**	27) **1**	28) **2**	29) **4**	30) **2**
31) **1**	32) **2**	33) **3**	34) **1**	35) **3**	36) **4**	37) **1**	38) **2**	39) **3**	40) **2**
41) **4**	42) **1**	43) **2**	44) **3**	45) **4**	46) **1**	47) **1**	48) **4**	49) **3**	50) **1**
51) **2**	52) **2**	53) **4**	54) **1**	55) **4**	56) **3**	57) **2**	58) **3**	59) **4**	60) **2**
61) **2**	62) **1**	63) **3**	64) **3**	65) **1**	66) **4**	67) **3**	68) **1**	69) **2**	70) **2**
71) **1**	72) **2**	73) **4**	74) **3**	75) **3**	76) **3**	77) **1**	78) **4**	79) **3**	80) **3**
81) **4**	82) **3**	83) **3**	84) **2**	85) **3**	86) **3**	87) **2**	88) **2**	89) **1**	90) **4**

PHYSICS

91) **1**	92) **1**	93) **1**	94) **4**	95) **3**	96) **3**	97) **1**	98) **2**	99) **2**	100) **4**
101) **2**	102) **3**	103) **3**	104) **2**	105) **4**	106) **2**	107) **3**	108) **2**	109) **2**	110) **2**
111) **2**	112) **1**	113) **1**	114) **4**	115) **4**	116) **3**	117) **3**	118) **1**	119) **3**	120) **1**
121) **4**	122) **4**	123) **1**	124) **1**	125) **1**	126) **2**	127) **3**	128) **4**	129) **2**	130) **2**
131) **3**	132) **4**	133) **2**	134) **3**	135) **4**					

CHEMISTRY

136) **1**	137) **3**	138) **4**	139) **3**	140) **4**	141) **3**	142) **4**	143) **3**	144) **4**	145) **3**
146) **2**	147) **1**	148) **2**	149) **1**	150) **3**	151) **1**	152) **2**	153) **1**	154) **2**	155) **2**
156) **2**	157) **1**	158) **3**	159) **4**	160) **4**	161) **2**	162) **1**	163) **4**	164) **2**	165) **2**
166) **1**	167) **1**	168) **1**	169) **2**	170) **4**	171) **4**	172) **3**	173) **3**	174) **3**	175) **4**
176) **4**	177) **3**	178) **2**	179) **4**	180) **4**					

91. $[AB]=\left[\frac{Vx}{\sqrt{x}}\right][x]=\left(ML^2T^{-2}\right)\left(L^{3/2}\right)$

$=\left[ML^{7/2}T^{-2}\right]$

92. $s_{AB}=u_{AB}t+\frac{1}{2}a_{AB}t^2$

t=10s

93. $\vec{V}_{RM}=4\cos 30\left(-\hat{k}\right)+4\sin\theta\left(-\hat{i}\right)$

$=\left(-2\sqrt{3}\hat{k}-2\hat{i}\right)\vec{V}_{RM}=\vec{V}_R-\vec{V}_M$

$\vec{V}_{RM}=\vec{V}_R-\vec{V}_M$

$\vec{V}_R=\left(-2\sqrt{3}\hat{k}-2\hat{i}\right)+4\hat{i}$

$\Rightarrow V_R=4ms^{-1}$

94. Area under v-t graph = displacement

95. $R=\frac{u^2}{g}=200m$

$H=\frac{u^2}{4g}=50\Rightarrow\vec{r}=\left(100\hat{i}+50\hat{j}\right)$

96. $F_{net}=5N,$ at 60^0 with initial velocity

97. $KE=\frac{1}{2}\times1\times(2-1)^2+\frac{1}{2}\times1(3+2)^2$

$=\frac{1}{2}\times1+\frac{1}{2}\times25=13J$

98. $P=MgV\Rightarrow V=\frac{10^4}{200\times10}ms^{-1}=5ms^{-1}$

$t=\frac{S}{V}=\frac{40}{5}=8s$

99. $\frac{dv}{dt}=9\Rightarrow dv=9dt\Rightarrow V=9t$

$a_c=a_t\Rightarrow\frac{V^2}{r}=a_t$

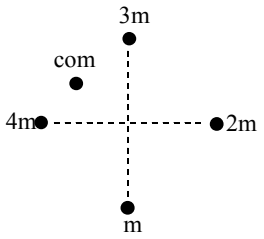
$\Rightarrow\frac{(9t)^2}{4}=9$

$t^2=\frac{9\times4}{9\times9}\Rightarrow t=\frac{2}{3}s$

100. $\Delta KE=\frac{1}{2}\frac{m_1m_2}{m_1+m_2}(u_1-u_2)^2\left(1-e^2\right)$

$e^2=\frac{1}{2}$

101.



102. $mgh_1=\frac{1}{2}mv^2\left(\frac{3}{2}\right)\&\,mgh_2=\frac{1}{2}mv^2$

103. $P:T:K=-2:-1:1$

$\Rightarrow\frac{PE}{\frac{1}{2}mv^2}=\frac{-2}{1}$

104. $\gamma=\frac{n_1C_{p_1}+n_2C_{p_2}}{n_1C_{v_1}+n_2Cv_2}=\frac{3\times\frac{8}{2}R+2\times\frac{5}{2}R}{3\times\frac{6R}{2}+2\times\frac{3R}{2}}$

$=\frac{17}{12}=1.42$

105. $PV=\frac{2}{3}E\Rightarrow E=\frac{3}{2}\times3\times10^6\times2$

$=9\times10^6J$

106. Angle of repose $\alpha=\tan^{-1}(\mu)$ remains same

107. $KE_{avg}=\frac{1}{4}mw^2A^2=\frac{1}{2}KE_{\max}$

Average speed $=\frac{4A\omega}{2\pi}=\frac{2}{\pi}V_{\max}$

108. $\frac{T^1}{T}=\sqrt{\frac{g}{g\left(1-\frac{\rho_l}{\rho_b}\right)}}\Rightarrow\frac{9}{1}=\frac{1}{\left(1-\frac{\rho_l}{\rho_b}\right)}\Rightarrow\frac{\rho_l}{\rho_b}=\frac{8}{9}$

109. $V=\sqrt{2gh}\Rightarrow\sqrt{2\times9.8\times10}=14ms^{-1}$

110. $\sigma=0.5\Rightarrow$ perfectly incompressible

$\Rightarrow B=$ infinity

111. Conceptual

112. $\left(\frac{dQ}{dt}\right)_{AB}=\left(\frac{dQ}{dt}\right)_{CB}$

$\frac{100-\theta}{l}=\frac{0-20}{l}\Rightarrow\theta=60^0C$

113. $\mu=\frac{\sin\left(\frac{A+D}{2}\right)}{\sin A/2}$ If μ increases D also increases

114. $e=e_0\sin(\omega t)$

at t=T/4, 'e' is maximum

at t=t/2, e is minimum

115. $M=\left(1+\frac{D}{f}\right)=\left(1+\frac{25}{10}\right)=3.5$

116. Conceptual

117. $\theta_P=\tan^{-1}(\mu)=60^0\Rightarrow i=60^0\ \&\ r=30^0$

118. Conceptual

119. Speed of $\bar{e}$ in EF increases but it remains same in M.F $\Rightarrow\lambda_1<\lambda_2$

120. $F\propto\frac{1}{r^2}\,but\ r\propto n^2$

121. $\frac{2N_0}{3}=N_0e^{-\lambda t_1}$

$\frac{N_0}{3}=N_0e^{\lambda t_2}\Rightarrow 2=e^{\lambda(t_2-t_1)}$

$\ln_2=\lambda(t_2-t_1)$

$(t_2-t_1)=\frac{\ln 2}{\lambda}=50\ days$

122. Emitter base junction is to be forward biased and base – collector junction is to be reverse biased

123. Conceptual

124. For damaged

oscillator $m\frac{d^2x}{dt}+b\frac{dx}{dt}+Kx=0$

For series LCR circuit

$L\left(\frac{d^2q}{dt^2}\right)+R\frac{dq}{dt}+\frac{1}{C}q=0$

$\Rightarrow m\to L,b\to R\ and\ K\to\frac{1}{C}$

125. Conceptual

126. $W=\Delta Vq$

127. $\frac{R+r}{R}=\frac{60}{50}\Rightarrow r=\left(\frac{10}{50}\right)6\Omega=1.2\Omega$

128. $\tan\delta^1=\frac{\tan\delta}{\cos D}$

129. $B_{inside}=0\ B_{outside}\neq 0$

130. $B=\frac{\mu_0}{2r}\left(\frac{qv}{2\pi r}\right)\&\ L=mvr$

$\Rightarrow B=\frac{\mu_0}{4\pi r^2}\left(\frac{L}{mr}\right)$

$B=\left(\frac{\mu_0 q}{4\pi r^3 m}\right)L$

131. Conceptual

132. $B=\frac{\mu_0 ni}{2r}=\frac{\mu_0 n}{2r}\left(\frac{VA}{\rho n2\pi r}\right)$

133. Conceptual

134. $n_1=n_2\Rightarrow\frac{P_1}{r_1}=\frac{P_2}{r_2}\Rightarrow\frac{P_1}{P_2}=\frac{1}{2}$

135. $V=\left(\frac{\Delta\lambda}{\lambda}\right)C=\frac{3\times10^8}{6\times10^3}=5\times10^4$